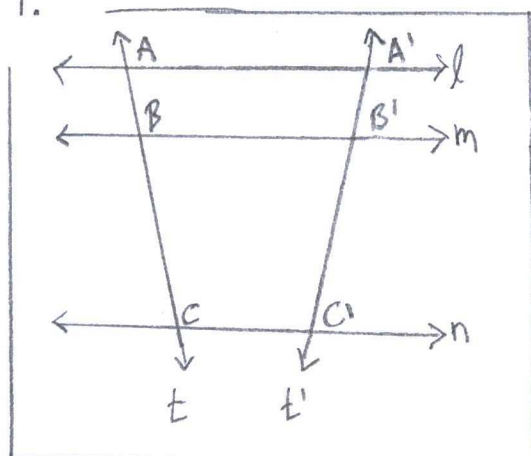


Chapter 5: History of Parallel Postulate

Major Exercises:

1.



"Parallel Projection Theorem"

a) (from exercise) "Opposite sides (and likewise, opposite angles) of a parallelogram are congruent to each other."

Proof: Hypothesis:

Evidence

1) Three parallel lines become reality: $\vec{l}$, $\vec{m}$, and $\vec{n}$.

Euclid's Postulate V

2) Two transversals $\vec{t}$ and $\vec{t'}$ intersect the parallels at a point s:

Proposition 2.1

3) A, B, C and A', B', C'

3) $AB/BC = A'B'/B'C'$

Parallel Projection Theorem

4) $\angle BCC' = \angle BB'C$
 $\leq 90^\circ$

RAH hypothesis

5) $BC/B'C' = BB'/CC'$

RAH result

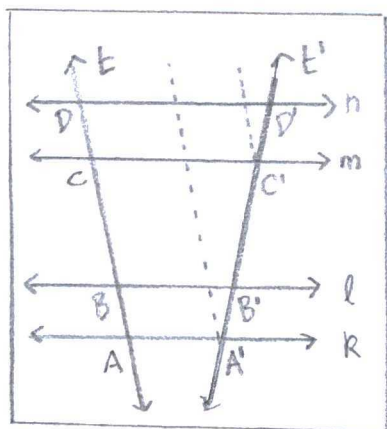
b) (from exercise)

"A parallelogram is a rectangle iff its diagonals are congruent, and in that case the diagonals bisect each other."

Proof	Hypothesis	Evidence
1) Three parallel lines become a reality: $\vec{l}, \vec{m},$ and $\vec{n}.$		Euclid's Postulate V
2) Two transversals $\vec{E}$ and $\vec{E'}$ intersect the parallels at points: A, B, C and A', B', C'		Proposition 2.1 Theorem
3) $\overline{AB}/\overline{BC} = \overline{A'B'}/\overline{B'C'}$		Parallel Projection Theorem
4) $\angle BCC' = \angle B'BC = 90^\circ$		RAA hypothesis
5) Segment $\overline{BC'} = \overline{B'C}$		
6) $\overline{BC'}$ crosses $\overline{B'C}$ at a point		RAA result

c) (from exercise) "A parallelogram has a circumscribed circle iff it is a rectangle"

Proof: Hypothesis	Evidence
1) Three parallel lines manifest into reality: $\vec{l}, \vec{m},$ and $\vec{n}.$	Euclid Postulate V
2) Two transversals $\vec{E}$ and $\vec{E'}$ intersect the parallels at points: A, B, C and A', B', C'	Proposition 2.1
3) $AB/BC = A'B'/B'C'$	Parallel Projection Theorem
4) $(\angle BCC' = \angle BB'C) = 90^\circ$	RAA hypothesis
5) Segments $BC' = B'C$	Euclid's Postulate II
6) $\angle CBB'$ and $\angle CC'B$	Euclid's Postulate III.22



"Corresponding angles..."

(from exercise) " $A'B' \cong C'D'$ "

Proof: Hypothesis	Evidence
1) Four parallel lines exist: $\vec{k}, \vec{l}, \vec{m}, \vec{n}$	Euclid's Postulate V
2) Transversals $\vec{E}$ and $\vec{E'}$	Proposition 2.1

Cut the parallels at
 A, B, C, D and A', B', C', D' ,
 respectively.

$$3) AB \cong CD \text{ when } l = m$$

$$4) AB/CD \cong A'B'/C'D'$$

$$5) A'B' \cong C'D'$$

RAA hypothesis
 Parallel Projection
 Theorem
 RAA outcome

3.

(from exercise) "Parallel Projection Theorem for the
 special case in which the ratio
 of lengths AB/BC is a rational
 number p/q "

Proof: Hypothesis

Evidence

1) When parallel lines
 become $\vec{l}, \vec{m},$ and $\vec{n}$

Euclid's Postulate V

2) Two transversals $\vec{E}$
 and $\vec{E'}$ intersect
 the parallels at
 points A, B, C and
 A', B', C'

Proposition 2.1

$$3) AB/BC = A'B'/B'C'$$

Parallel Projection
 Theorem

4) A segment exists

Archimedes Axiom

$$\text{so } AB \cong p \cdot XY$$

$$5) BC \cong q \cdot XY$$

$$6) p/q = AB/BC$$

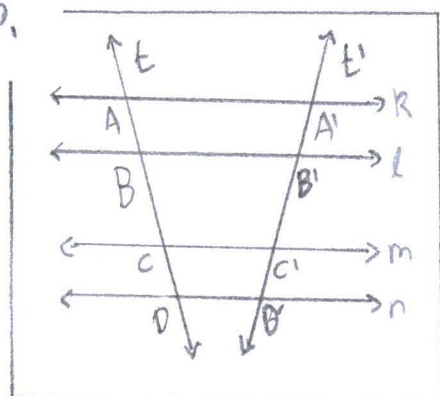
$$= n(p+q)$$

when p and q

are even

(from exercise) " AB/BC is an irrational number"

5.



"Irrational ratios"

Proof: — Hypothesis

1) Three lines are parallel to $\vec{R}$, specifically $\vec{L}$, $\vec{m}$ and $\vec{n}$.

2) $\vec{E}$ and $\vec{E'}$ intersect each parallel at points A, B, C and A', B', C' .

$$3) AB/BD = A'B'/B'D'$$

$$4) BD \cong p \cdot CB$$

$$5) BA = \frac{BD}{q} = \frac{p \cdot CB}{q}$$

Archimedes Axiom

Archimedes Axiom

Evidence

Euclid's Parallel Postulate

Proposition 2.1

Parallel Projection Theorem

Archimedes Axiom

Archimedes Axiom

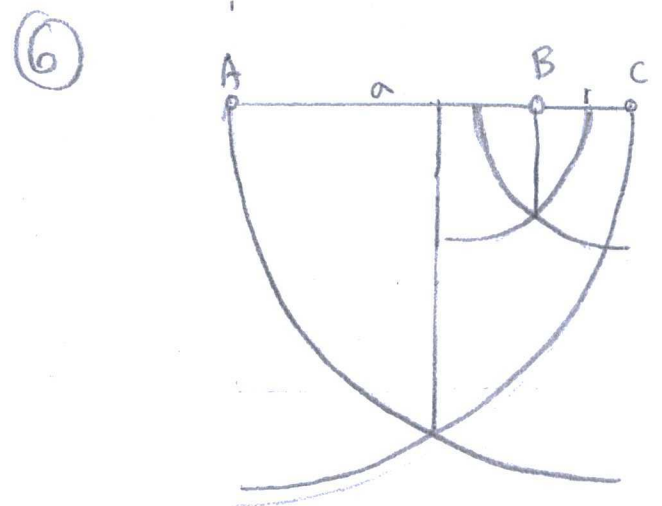
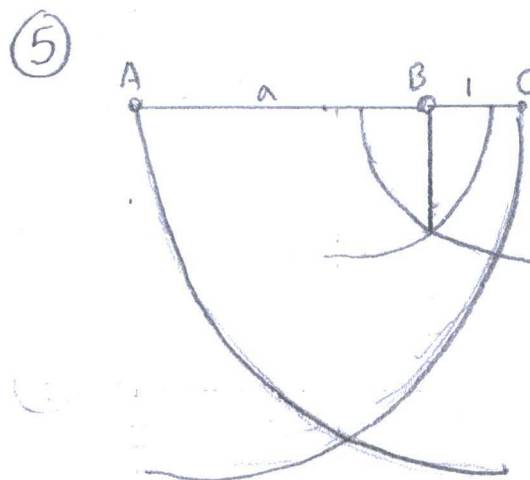
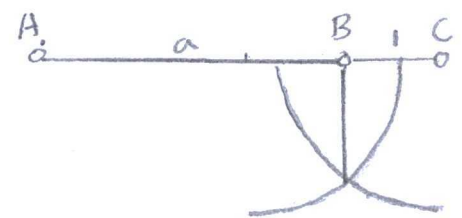
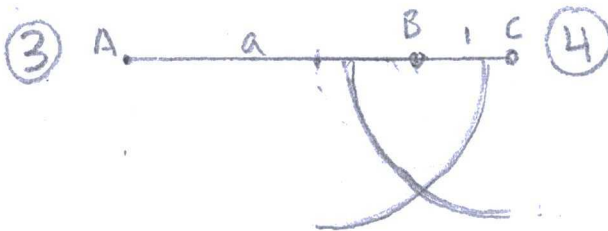
7) At an irrational
ratio less than p/q ,
 $B \times D \times A$ is necessary.

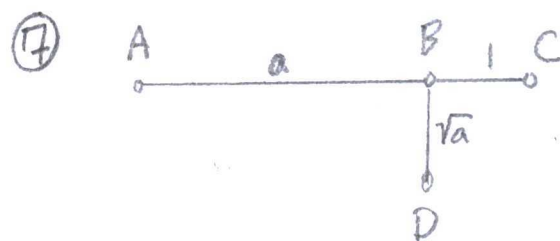
RAA hypothesis (False)

8) So $A \times B \times C \times D$ and
 $p/q = AB/BD$ remains
irrational.

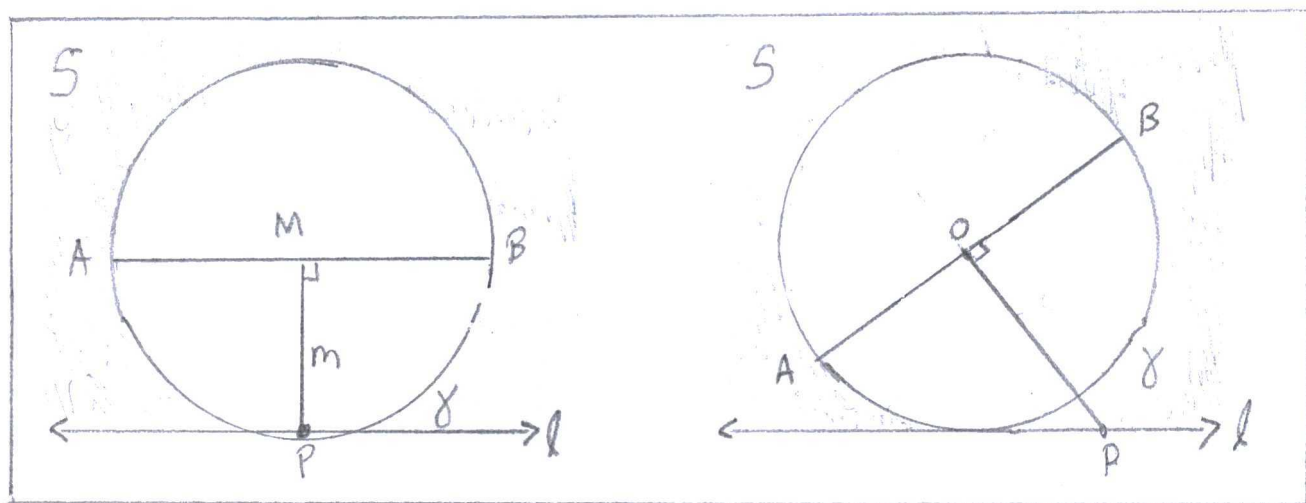
6.

By straight edge and compass, segment length $\sqrt{a}$





7.



"Same side... of l "

Proof: Hypothesis

- 1) A line, a single $\vec{l}$
- 2) Two points A and B are not on $\vec{l}$, but same side
- 3) Point M is the midpoint on AB
- 4) A perpendicular bisector m meets line $\vec{l}$ at point P

Case 1: $AB \parallel l$

Evidence

(pg 12) "Definition-line"

Axiom B-4

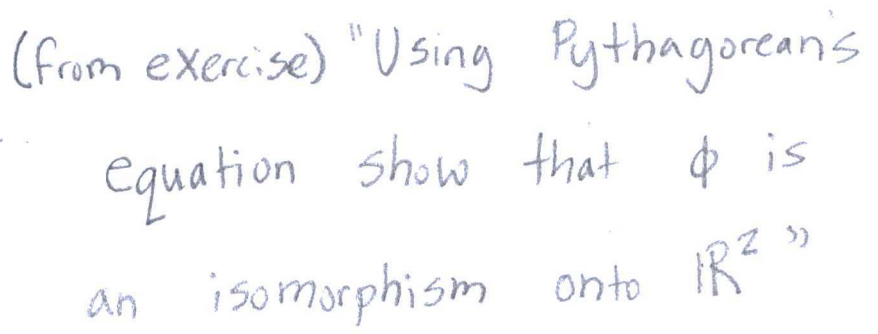
Proposition 4. (midpoint)

Proposition 4. (bisector)

Euclid's Postulate III

Euclid Postulate III

(from exercise) "Given any line l , two points A and B not on l are on the same side S of l if and only if they lie on a circle contained in S "



Note : $\phi(0) = (0, 0)$
 $\phi(P) = (\pm x, \pm y)$
 $\phi(P') = (+x, 0)$
 $\phi(P^n) = (0, y)$

Pythagoreas' Equation relates to the distance formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{a^2 + b^2}$$

$$= c$$

$$c^2 = a^2 + b^2$$